Verification of Open Interactive Markov Chains

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Overview

1. Significance of Interactive Markov Chains
2. Setting Up Basic Terminology
3. Compositional Modelling Using IMCs
4. Controller-Environment Games
5. Discretization
6. Conclusion

Introducing Interactive Markov Chains

Why do we need yet another model?

- Consider an autonomously driving car; to model it we need to model

Random Events

1. Behaviour of pedestrians
2. Component Failure
3. Traffic Lights Behaviour

Controlled Decisions

1. Choosing a route
2. Overtaking another vehicle
3. Brakes, Honking

- We need a model to blend the stochasticity of the process while accommodating controlled decisions

Setting Up Notations

Defining Interactive Markov Chains (1)

An IMC is represented as a tuple $\mathcal{C} = (S, \mathbb{A}\mathrm{ct}^{\tau}, \hookrightarrow, \rightsquigarrow, s_0)$ where S is a finite set of *states*, $\mathbb{A}\mathrm{ct}^{\tau}$ is a finite set of *actions* containing a designated *internal action* τ, $s_0 \in S$ is an *initial* state,

- $\hookrightarrow \subseteq S \times \mathbb{A}\mathrm{ct}^{\tau} \times S$ is an *interactive transition* relation, and
- $\rightsquigarrow \subseteq S \times \mathbb{R}_{>0} \times S$ is a *Markovian transition* relation

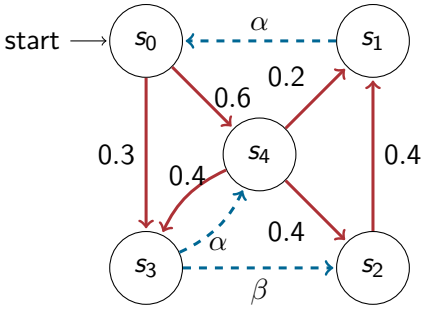
Defining Interactive Markov Chains (2)

- Elements of $\mathbb{A}\mathrm{ct} := \mathbb{A}\mathrm{ct}^{\tau} \setminus \{\tau\}$ are called *external* actions
- **Open IMCs**: external actions controlled by environment
  Closed IMCs: external actions controlled by other system components
- $\mathbf{E} : S \to \mathbb{R}_{\geq 0}$ assigns to a state the sum of rates of all outgoing Markov transitions

IMC Example

Consider an IMC with states $S = \{s_0, s_1, s_2, s_3, s_4\}$ and:

- Interactive transitions: dashed arrows
- Markovian transitions: solid arrows



Markovian Transitions

- **Total exit rate function** $E : S \to \mathbb{R}_{\geq 0}$ such that $E(s) = \Sigma_{s \overset{\lambda}{\rightsquigarrow} t} \lambda$
- **Probability Matrix** $P(s,t) = \lambda / E(s)$if $s \overset{\lambda}{\rightsquigarrow} t$, else 0

Compositional Modelling Using IMCs

- **Parallel Composition**: Syncing various components of the system using shared actions
- **Hiding Operator**: Controls action visibility and manages system abstraction
- **Maximal Progress Assumption**: Internal transitions happen instantaneously and precede delay transitions whereas external transitions don't

Parallel Composition (1)

Given **IMCs**

$$\mathcal{C}_1 = (S_1, \mathbb{A}\mathrm{ct}_1^\tau, \hookrightarrow_1, \rightsquigarrow_1, s_{01}) \text{ and } \mathcal{C}_2 = (S_2, \mathbb{A}\mathrm{ct}_2^\tau, \hookrightarrow_2, \rightsquigarrow_2, s_{02}),$$

and **synchronisation alphabet** $A \subseteq \mathbb{A}ct_1 \cap \mathbb{A}ct_2$, their parallel composition $\mathcal{C}_1 \,||_A\, \mathcal{C}_2$ is given as

$$\mathcal{C} = (S_1 \times S_2, \mathbb{A}\mathrm{ct}_1^\tau \times \mathbb{A}\mathrm{ct}_2^\tau, \hookrightarrow, \rightsquigarrow, (s_{01}, s_{02}))$$

Parallel Composition (2)

- $s_1 \overset{a}{\hookrightarrow} s_1'$ and $s_2 \overset{a}{\hookrightarrow} s_2'$ and $a \in A$ implies $(s_1, s_2) \overset{a}{\hookrightarrow} (s_1', s_2')$
- $s_1 \overset{a}{\hookrightarrow} s_1'$ and $a \notin A$ implies $(s_1, s_2) \overset{a}{\hookrightarrow} (s_1', s_2)$ for each $s_2 \in S_2$
- $s_2 \overset{a}{\hookrightarrow} s_2'$ and $a \notin A$ implies $(s_1, s_2) \overset{a}{\hookrightarrow} (s_1, s_2')$ for each $s_1 \in S_1$
- $s_1 \overset{\lambda}{\rightarrow} s_1'$ implies $(s_1, s_2) \overset{\lambda}{\rightsquigarrow} (s_1', s_2)$ for each $s_1 \in S_1$
- $s_2 \overset{\lambda}{\rightarrow} s_2'$ implies $(s_1, s_2) \overset{\lambda}{\rightsquigarrow} (s_1, s_2')$ for each $s_2 \in S_2$

Parallel Composition (3)

- Example: Consider two IMCs:
  - $\mathcal{C}_1$: S:$\{s_0, s_1, s_2\}$ $\mathbb{A}$ct:{(I) α =request$(s_0 - s_1)$, γ =send$(s_0 - s_2)$, (M) $\lambda = 2.0$ $(s_1 - s_2)$}
  - $\mathcal{C}_2$: S:$\{t_0, t_1, t_2\}$ $\mathbb{A}$ct:{(I) α =request$(t_0 - t_1)$, β =wait$(t_0 - t_2)$, (M) $\mu = 3.0$ $(t_1 - t_2)$}
- For $A = \{\alpha\}$, in $\mathcal{C}_1 || _A \mathcal{C}_2$

$$(s_0, t_0) \xrightarrow{\alpha} (s_1, t_1),\ (s_0, t_i) \xrightarrow{\gamma} (s_2, t_i),\ (s_i, t_0) \xrightarrow{\beta} (s_2, t_i),$$
$$(s_1, t_i) \xrightarrow{\lambda} (s_2, t_i),\ (s_i, t_1) \xrightarrow{\mu} (s_i, t_2)$$

Hiding (1)

Given IMC $\mathcal{C} = (S, \mathbb{A}\mathrm{ct}^\tau, \hookrightarrow, \rightsquigarrow, s_0)$, a hidden alphabet $A \subseteq \mathbb{A}ct$, the hiding $\mathcal{C}\backslash A$ is $\mathcal{C}' = (S, \mathbb{A}\mathrm{ct}^\tau \setminus A, \hookrightarrow', \rightsquigarrow, s_0)$ where $\hookrightarrow'$ is the smallest relation satisfying

- $a \in A$ implies $s \overset{\tau}{\hookrightarrow} s'$, and,
- $a \notin A$ $s \overset{a}{\hookrightarrow}' s'$

for each $s \overset{a}{\hookrightarrow} s'$

Hiding (2)

- Example: Consider an IMC $\mathcal{C}$ with states $S = \{s_0, s_1, s_2, s_3, s_4\}$ and the following actions:

$$\{[I]\ \alpha \to \text{request}(s_0 - s_1),\ \beta \to \text{process}(s_1 - s_2), [M]\ \lambda = 3.0\ (s_2 - s_0)\}$$

- For $a = \{\beta\}$, we get IMC $\mathcal{C}' = (S, \mathbb{A}ct^{\beta}, \hookrightarrow', \rightsquigarrow, s_0)$

Scheduler (1)

- Needed to resolve non-determinism amongst internal transitions, manage timing decisions and ensure determinism for analysis
- for IMC $\mathcal{C} = (S, \mathbb{A}\text{ct}^{\tau}, \hookrightarrow, \rightsquigarrow, s_0)$, scheduler function $\sigma : (S \times \mathbb{R}_{\geq 0})^* \times S \to S$ such that
  - σ is a measurable function
  - for each history $\mathfrak{h} = s_0 t_0 s_1 ... s_n$ with $\text{succ}_{\tau}(s_n) \neq \phi$ we have: $\sigma \in \text{succ}_{\tau}(s_n)$
- $\mathfrak{S}(\mathcal{C})$ is the set of all schedulers of $\mathcal{C}$

Compositional Verification Using IMCs

Time Bounded Reachability

- Can the system reach a goal state in the defined time limit?
- Given goal states $G \subseteq S$ and time-bound $T \in \mathbb{R}_{\geq 0}$, the value of time-bounded reachability is

$$sup_{\sigma \in \mathfrak{S}(\mathcal{C})} \mathcal{P}^{\sigma}_{\mathcal{C}}[\Diamond^{\leq T} G],$$

where $\mathcal{P}^{\sigma}_{\mathcal{C}}[\Diamond^{\leq T} G]$ is the probability of reaching G before T

Compositional Verification (1)

- How well does a component perform (wrt time-bounded reachability) when working in parallel and with an unknown environment?
- Equivalent to finding a scheduler that maximises the chances of reaching G before T irrespective of the environment E that $\mathcal{C}$ is composed with

Compositional Verification (2)

- Environment: IMC E such that $\mathbb{A}\mathrm{ct}_E = \mathbb{A}\mathrm{ct}_{\mathcal{C}}$
- Consider scheduler σ for $\mathcal{C}$ and π for resolution of non-determinism within E and that arising from syncing $\mathcal{C}$ and E
- What we are looking for:

  $\sup_{\sigma} \inf_{E,\pi} \mathcal{P}[G \text{ is reached in composition of } \mathcal{C} \text{ and } E \text{ before } T \text{ using } \sigma \text{ and } \pi]$

Compositional Verification (3)

- For a given environment E with the same $\mathbb{A}\mathrm{ct}^{\tau}$, we give the composition:

$$\mathcal{C}(E) \;=\; (\mathcal{C} \;||_{\mathbb{A}ct}\; E)\backslash\mathbb{A}ct$$

- Consider σ scheduler of $\mathcal{C}$ and π a scheduler of $\mathcal{C}(E)$; π respects σ if for every history $\mathfrak{h}$ of $\mathcal{C}(E)$:
  - $\pi(\mathfrak{h}) = (c, e)$ where $c_n \overset{a}{\hookrightarrow} c$ and $e_n \overset{a}{\hookrightarrow} e$ (synchronisation resolved)
  - $\pi(\mathfrak{h}) = (c_n, e)$ where $e_n \overset{\tau}{\hookrightarrow} e$ (π governs choice of moves in E)
  - $\pi(\mathfrak{h}) = (\sigma(\mathfrak{h}_{\mathcal{C}}), e_n)$ where $e_n \overset{\tau}{\hookrightarrow} e$ where $h_{\mathcal{C}}$ is $\mathfrak{h}$ projecting over the states of E (π chooses moves in $\mathcal{C}$ as per σ)

Compositonal Verification (4)

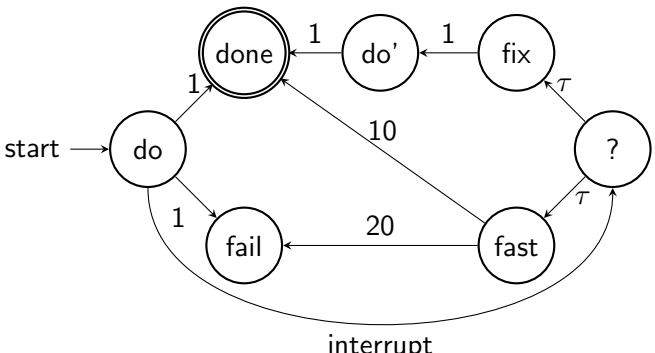
In $\mathcal{C}(E)$, the value of time-bounded reachability is

$$R^{\mathcal{C}(E)}_{T,G} = \sup_{\sigma \in \mathfrak{S}(\mathcal{C})} \inf_{\substack{E \in \mathrm{ENV} \\ \pi \in \mathfrak{S}(\mathcal{C}(E))}} P^{\pi}_{\mathcal{C}(E)}[\Diamond^{\leq T} G_E]$$

where $G_E = G \times S_E$ and ENV is the set of all IMCs with alphabet $\mathbb{A}\mathrm{ct}$

Compositional Verification Example

- $T = 2$, $G = \{\text{done}\}$
- At ?, σ chooses between fast & fix
- $R^{\mathcal{C}(\mathsf{E})}_{T,G} \approx 0.47$ and σ_{opt} chooses fix if $t \geq 1.2\text{min}$
- Most adversarial E causes *interrupt* after 0.8 minutes from *start*



Assumptions

1. Each cycle contains a Markovian transition
   - Implies that the probability of Zeno behaviour is zero
     - Zeno behaviour: a run where infinite transitions occur within a finite time
   - A compositional property (preserved by hiding and parallel composition)
2. Internal and external actions are not enabled at the same time, ie, for each state, $\text{succ}_\tau(s) = \phi$ or $\text{succ}_e(s) = \phi$
   - NOT compositional (not preserved in hiding)
   - Causes internal transitions to be executed before any external actions are taken into account

What do we want?

- Let $\varepsilon > 0$ be an approximation bound and $\mathcal{C} = (S, \mathbb{A}\mathrm{ct}^\tau, \hookrightarrow, \rightsquigarrow, s_0)$ be an IMC satisfying Assumptions 1 and 2. $R^{\mathcal{C}(\mathcal{E})}_{T,G}$ can be approximated up to ε and ε-optimal scheduler can be computed in time

$$\mathcal{O}(|S|^2 \cdot \tfrac{(\lambda T)^2}{\varepsilon}),$$

where λ is the maximal rate of $\mathcal{C}$

Controller Environment Game

Stochastic Games

- Directed graph $G = (V, E)$ where
  - $V = V_1 \cup V_2$ and $V_1 \cap V_2 = \phi$ $V_1 \rightarrow$Max player, $V_2 \rightarrow$ Min player
  - Each player decides on which transition to take in their corresponding set of vertices
  - Max player tries to maximise the chances of reaching goal vertices and Min player tries to minimise these chances

Controller-Environment Games

- Two-player game to combine the stochastic and non-deterministic flow of time with players **con** and **env**
- **con**- chooses which internal transition to take in the states which have them
  env- chooses which external transition in states that have them to take or a delay $t_e \in \mathbb{R}_{>0}$

Characteristics of CE Games

- Due to assumption 2, $\mathcal{G}$ is a turn-based game
- **con** tries to reach a subset G of states before time T elapses, and **env** tries to choose delay to ensure that **con** doesn't reach goal states in time and maximise the risk of less favorable Markovian transitions

How does $\mathcal{G}$ proceed?

- **con** decides the v_{i+1} when $v_i \in V_{\Box}$
- **env** decides the v_{i+1} when $v_i \in V_{\Diamond}$
- v_{i+1} is decided randomly when $v_i \in V_{\circ}$

A Run of $\mathcal{G}$ (1)

- A run of $\mathcal{G}$ is an infinite sequence $s_0\, t_0\, s_1\, t_1\, \ldots$
- Based on the history $\mathfrak{h}$, next move is chosen
  - $\mathrm{succ}_\tau(s_n) \neq \emptyset \rightarrow$ **con** chooses $s_r \in \mathrm{succ}_\tau(s_n)$
  - Otherwise, **env** chooses $s_e \in \mathrm{succ}_e(s_n)$ or delay $t_e \in \mathbb{R}_{>0}$
  - Markovian transitions (if available) resolved by choosing state s_M according to $P(s_n, \cdot)$ and randomly sampling t_M according to exponential distribution with rate $E(s_n)$

A Run of $\mathcal{G}$ (2)

- Next waiting time t_n and state s_{n+1} are given by
  - If $\mathrm{succ}_\tau(s_n) \neq \emptyset$ and s_r was chosen, then $t_n = 0$ and $s_{n+1} = s_r$
  - If s_e was chosen, then $t_n = 0$ and $s_{n+1} = s_e$
  - If t_e was chosen,
    - $\mathrm{succ}_\mathrm{M}(s_n) = \emptyset$, then $t_n = t_e$ and $s_{n+1} = s_n$
    - If $t_e \leq t_M$, then $t_n = t_e$ and $s_{n+1} = s_n$
    - If $t_M < t_e$, then $t_n = t_M$ and $s_{n+1} = s_M$

Value of $\mathcal{G}$

- **con** $\rightarrow$ $\sigma : (S \times \mathbb{R}_{\geq 0})^* \times S \rightarrow S$

  env $\rightarrow$ $\pi : (S \times \mathbb{R}_{\geq 0})^* \times S \rightarrow S \cup \mathbb{R}_{>0}$

- Value of $\mathcal{G}$ is given by

$$R^{\mathcal{G}}_{T,G} = \sup_{\sigma \in \Sigma} \inf_{\pi \in \Pi} \ \mathcal{P}^{\sigma,\pi}_{\mathcal{G}}[\Diamond^{\leq T} G]$$

- **Proposition**: $R^{\mathcal{C}(E)}_{T,G} = R^{\mathcal{G}}_{T,G}$

Proving Equivalence of Values (1)

- We start with the inequality $R^{\mathcal{C}}_{T,G} \geq R^{\mathcal{G}}_{T,G}$
- The purpose of the environment E is to choose delays of external actions
- The only difference is that environment E "chooses" the delays randomly unlike deterministic π of *env*

Proving Equivalence of Values (2)

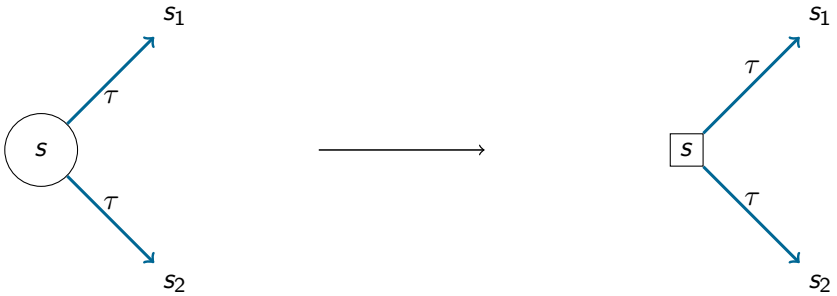
- Now we consider $R^{\mathcal{C}}_{T,G} \leq R^{\mathcal{G}}_{T,G}$
- Every π' of *env* can be approximated a suitable environment with scheduler π
- Every deterministic delay t chosen by *env* randomly corresponds to Erlang distribution centered around t

Discretization

Discretization (1)

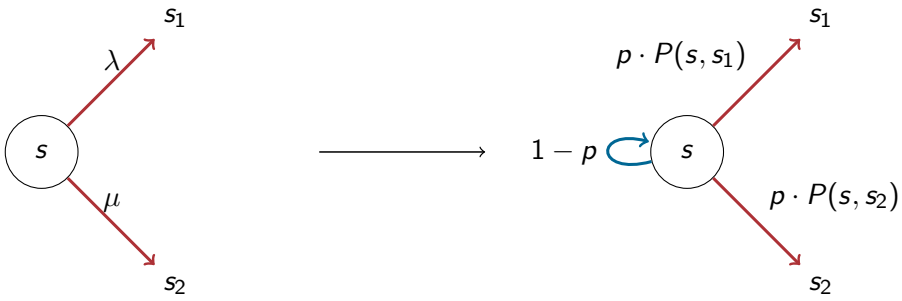
- Conversion of continuous CE game into discrete time model
- A stochastic game Δ played on $(V, \mapsto)$ such that $V = V_{\Box} \uplus V_{\Diamond} \uplus V_{\circ}$
- Player $\Box \rightarrow V_{\Box}, \sigma \in \Sigma_{\Delta}$
  Player $\Diamond \rightarrow V_{\Diamond}, \pi \in \Pi_{\Delta}$
  $V_{\circ} \rightarrow$: v_{i+1} chosen randomly as per $Prob(v_i)$

Discretising States in IMC (1)



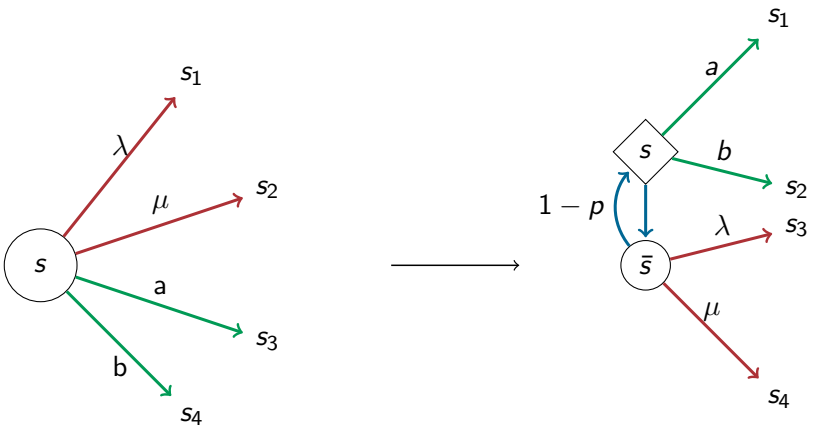
Discretising $S_{\square}$

Discretising States in IMC (2)



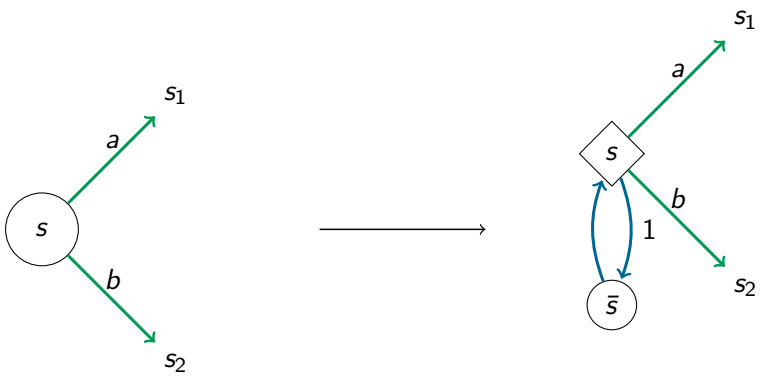
Discretising vertices with only Markovian Transitions

Discretising States in IMC (3)



Discretising vertices with both Markovian and External Actions

Discretising States in IMC (4)



Discretising vertices with External Actions only

Discretization (2)

- Fix $\delta > 0$ time step such that $\delta = \frac{T}{N}$ for some $N \in \mathbb{N}$
- Step from $V_\Box \cup V_\Diamond$ take 0 time and steps from $V_\circ$ take δ time
- Value of game Δ is

$$R^{\Delta}_{T,G} = \sup_{\sigma \in \Sigma_\Delta} \inf_{\pi \in \Pi_\Delta} \mathcal{P}^{\sigma,\pi}_{\Delta}[\Diamond^{\#_\circ \leq N} G]$$

where $\mathcal{P}^{\sigma,\pi}_{\Delta}[\Diamond^{\#_\circ \leq N} G]$ is the probability of all runs of Δ that reach G before taking more than N steps from vertices of $V_\circ$

Discretization (3)

- A strategy is *counting* if only the last vertex and the current count $\#_\circ$; can be represented as a fxn : $V \times \{0, ...N\} \to V$
- There are counting strategies optimal in $R^{\Delta}_{T,G}$ and they can be computed with $R^{\Delta}_{T,G}$ in $\mathcal{O}(N|V|^2)$

Error Bound [Estimating $R^{\mathcal{G}}_{T,G}$] (1)

Proposition

For every approximation bound $\varepsilon > 0$ and $\delta \leq \frac{\varepsilon}{\lambda^2 T}$ where $\lambda = \max_{s \in S} \mathbf{E}(s)$ the value $R^{\Delta}_{T,G}$ induced by δ satisfies

$$R^{\Delta}_{T,G} \;\leq\; R^{\mathcal{G}}_{T,G} \;\leq\; R^{\Delta}_{T,G} + \varepsilon$$

Error Bound (2.1)

- Δ restricts possible behaviours of the CE game:
  - Only one Markovian transition allowed in any interval of δ
  - t_e must be a multiple of δ
- None of these affects the value of the game

Error Bound(2.2.a)

- Probability of two or more Markovian transitions occurring in interval $[0, \delta]$ is bounded by $\frac{(\lambda\delta)^2}{2}$ $\rightarrow$ the probability of taking multiple Markovian transitions in a discrete time-step is $\leq \varepsilon$

Error Bound(2.2.b)

- Assuming at most one Markovian transition in $[0, \delta]$, the decision of when to choose external transitions is reduced to minimising a linear function on $[0, \delta]$
  - min either at δ or at 0
  - Optimal choice for **env** to take transition at beginning of time step or wait until for time δ

Error Bound [Estimating $R^{\mathcal{C}}_{T,G}$] (3)

- Let $\varepsilon > 0$, Δ be a corresponding discrete game, and σ be induced by an optimal counting strategy in Δ, then
$$R^{\mathcal{C}(E)}_{T,G} \leq \inf_{\substack{E \in \mathsf{ENV} \\ \pi \in \mathfrak{S}(\mathcal{C}(E),\sigma)}} P^{\pi}_{\mathcal{C}(E)}[\Diamond^{\leq T} G_E] + \varepsilon$$
- This together with the complexity result of computing counting strategies finishes the proof of the central theorem.

Conclusion and Discussion

- Computation of maximal timed bounded reachability for IMC operating in an unknown IMC environment
- Resulting stochastic two-player game has the same extremal values as a CE-game
- Approximated up to a given precision by discretization and the resulting control strategy translated back to a scheduler of the IMC achieving the bound.

Future Work and Applications

- Robustness wrt rate changes for IMCs remains unexplored
- More efficient abstraction while preserving important behaviour
- Modelling cyber-physical systems using IMCs

The End